

Two-way fixed effects estimators with heterogeneous treatment effects*

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Abstract

Linear regressions with period and group fixed effects are widely used to estimate treatment effects. We show that they estimate weighted sums of the average treatment effects (ATE) in each group and period, with weights that may be negative. Due to the negative weights, the linear regression coefficient may for instance be negative while all the ATEs are positive. We propose another estimator that solves this issue. In the two applications we revisit, it is significantly different from the linear regression estimator.

Keywords: linear regressions, fixed effects, heterogeneous treatment effects, difference-in-differences.

JEL Codes: C21, C23

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Our paper is related to the DID literature. Our main result generalizes Theorem 1 in de Chaisemartin and D’Haultfoeuille (2018). When the data has two groups and two periods, the Wald-DID estimand considered therein is equal to β_{fe} and β_{fd} . Our results on β_{fe} and β_{fd} are thus extensions of that theorem to the case with multiple periods and groups.³ Moreover, our DID_M estimator is related to the Wald-TC estimator with many groups and periods proposed in de Chaisemartin and D’Haultfoeuille (2018), and to the multi-period DID estimator proposed by Imai and Kim (2018). In Section 4, we explain the differences between those three estimators.

More recently, Borusyak and Jaravel (2017), Abraham and Sun (2018), Athey and Imbens (2018), Callaway and Sant’Anna (2018), and Goodman-Bacon (2018) study the special case of staggered adoption designs, where the treatment of a group is weakly increasing over time. Those papers derive some important results specific to that design that we do not consider here. Still, some of the results in those papers are related to ours, and we describe precisely those connections later in the paper. The most important dimension on which our paper differs from those is that our results apply to any two-way fixed effects regressions, not only to those with staggered adoption. In our survey of the AER papers estimating two-way fixed effects regressions, less than 10% have a staggered adoption design. This suggests that while staggered adoptions are an important research design, they may account for a relatively small minority of the applications where two-way fixed effects regressions have been used.

The paper is organized as follows. Section 2 introduces the set-up. Section 3 presents our decomposition results. Section 4 introduces our alternative estimator. Section 5 briefly describes some of the extensions covered in our Web Appendix. Section 6 presents our survey of the articles published in the AER, and our two empirical applications.

2 Set up

One considers observations that can be divided into G groups and T periods. For every $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $N_{g,t}$ denote the number of observations in group g at period t , and let $N = \sum_{g,t} N_{g,t}$ be the total number of observations. The data may be an individual-level panel or repeated cross-section data set where groups are, say, individuals’ county of birth. The data could also be a cross-section where cohort of birth plays the role of time. For instance, Duflo (2001) compares the schooling of different cohorts in Indonesia, some of which were exposed to a school construction program. It is also possible that for all (g, t) , $N_{g,t} = 1$, e.g. a group is one individual or firm. All of the above are special cases of the data structure we consider.

One is interested in measuring the effect of a treatment on some outcome. Throughout the paper

³In fact, a preliminary version of our main result appeared in a working paper version of de Chaisemartin and D’Haultfoeuille (2018) (see Theorems S1 and S2 in de Chaisemartin and D’Haultfoeuille, 2015).

Here is some intuition as to why one weight is negative in this example. It follows from Equation (7) in the proof of Theorem 1 (see also Theorem 1 in Goodman-Bacon, 2018) that in this simple example, $\beta_{fe} = (\text{DID}_1 + \text{DID}_2)/2$, with

$$\begin{aligned}\text{DID}_1 &= E(Y_{2,2}) - E(Y_{2,1}) - (E(Y_{1,2}) - E(Y_{1,1})), \\ \text{DID}_2 &= E(Y_{1,3}) - E(Y_{1,2}) - (E(Y_{2,3}) - E(Y_{2,2})).\end{aligned}$$

The first DID compares the evolution of the mean outcome from period 1 to 2 in group 2 and in group 1. The second one compares the evolution of the mean outcome from period 2 to 3 in group 1 and in group 2. The control group in the second DID, group 2, is treated both in the pre and in the post period. Therefore, under the common trends assumption, it follows from Lemma 1 in Appendix A (a similar result appears in Lemma 1 of de Chaisemartin (2011) and in Equation (13) of Goodman-Bacon (2018)) that $\text{DID}_1 = E[\Delta_{2,2}]$, but

$$\text{DID}_2 = E[\Delta_{1,3}] - (E[\Delta_{2,3}] - E[\Delta_{2,2}]).$$

DID_2 is equal to the ATE in group 1 period 3, minus the change in group 2's ATE between periods 2 and 3. Intuitively, the mean outcome of groups 1 and 2 may follow different trends from period 2 to 3 either because group 1 becomes treated, or because group 2's ATE changes. The intuition that negative weights arise because $\hat{\beta}_{fe}$ uses treated observations as controls also appears in Borusyak and Jaravel (2017).

We now generalize the previous illustration by characterizing the (g, t) cells whose ATEs are weighted negatively by β_{fe} .

Proposition 1 *Suppose that Assumption 1 holds and for all $t \geq 2$ $N_{g,t}/N_{g,t-1}$ does not vary across g . Then, for all (g, t, t') such that $D_{g,t} = D_{g,t'} = 1$, $D_{.,t} > D_{.,t'}$ implies $w_{g,t} < w_{g,t'}$. Similarly, for all (g, g', t) such that $D_{g,t} = D_{g',t} = 1$, $D_{g,.} > D_{g',.}$ implies $w_{g,t} < w_{g',t}$.*

Proposition 1 shows that β_{fe} is more likely to assign a negative weight to periods where a large fraction of groups are treated, and to groups treated for many periods. Then, negative weights are a concern when treatment effects differ between periods with many versus few treated groups, or between groups treated for many versus few periods.

Proposition 1 has interesting implications in staggered adoption designs, a special case of sharp designs defined as follows.

Assumption 6 (*Staggered adoption designs*) *For all g , $D_{g,t} \geq D_{g,t-1}$ for all $t \geq 2$.*

Assumption 6 is satisfied in applications where groups adopt a treatment at heterogeneous dates (see e.g. Athey and Stern, 2002). In that design, Borusyak and Jaravel (2017) show that β_{fe} is more likely to assign a negative weight to treatment effects at the last periods of the panel.

This result is a special case of Proposition 1: in staggered adoption designs, $D_{.,t}$ is increasing in t , so Proposition 1 implies that $w_{g,t}$ is decreasing in t .¹⁰ Proposition 1 also implies that in that design, groups that adopt the treatment earlier are more likely to receive some negative weights.

Finally, in staggered adoption designs, Athey and Imbens (2018) derive a decomposition of β_{fe} that resembles to, but differs from, that in Theorem 1. They derive their decomposition under the assumption that the dates at which each group starts receiving the treatment are randomly assigned, while we derive ours under a common trends assumption.

3.2 Robustness to heterogeneous treatment effects

Theorem 1 shows that in sharp designs with many groups and periods, $\hat{\beta}_{fe}$ may be a misleading measure of the treatment effect under the standard common trends assumption, if the treatment effect is heterogeneous across groups and time periods. In the corollary below, we propose two robustness measures that can be used to assess how serious that concern is.

Those robustness measures are defined conditional on $\mathbf{D}$, the vector stacking together the treatments of all the (g, t) cells. Specifically, for all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | \mathbf{D})$ denote the ATE in cell (g, t) conditional on $\mathbf{D}$,¹¹ let $\tilde{\Delta}^{TR} = E(\Delta^{TR} | \mathbf{D})$ denote the ATT conditional on $\mathbf{D}$, and let $\tilde{\beta}_{fe} = E(\hat{\beta}_{fe} | \mathbf{D})$. The first measure we consider is the minimal value of the standard deviation of the $\tilde{\Delta}_{g,t}$ s under which one could have that $\tilde{\beta}_{fe}$ is of a different sign than $\tilde{\Delta}^{TR}$. Therefore, this summary measure applies to $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$, rather than β_{fe} and δ^{TR} , the unconditional expectations of $\hat{\beta}_{fe}$ and Δ^{TR} on which we have focused so far. However, one can show that when G , the number of groups, goes to infinity, $\tilde{\beta}_{fe} - \beta_{fe}$ and $\tilde{\Delta}^{TR} - \delta^{TR}$ both converge to 0. So if the number of groups is large, $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$ should not differ much from β_{fe} and δ^{TR} , and our robustness measure “almost” applies to β_{fe} and δ^{TR} .

Let

$$\sigma(\tilde{\Delta}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (\tilde{\Delta}_{g,t} - \tilde{\Delta}^{TR})^2 \right)^{1/2},$$

$$\sigma(\mathbf{w}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1)^2 \right)^{1/2}.$$

¹⁰Borusyak and Jaravel (2017) assume that the treatment effect of cell (g, t) only depends on the number of periods since group g has started receiving the treatment, whereas Proposition 1 does not rely on that assumption.

¹¹ $\tilde{\Delta}_{g,t}$ may differ from $E(\Delta_{g,t})$. To see this, let us consider a simple example where $T = 2$. Then, under Assumption 3, one has $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | D_{g,1}, D_{g,2})$. One may for instance have $E(\Delta_{g,1} | D_{g,1} = 0, D_{g,2} = 0) < E(\Delta_{g,1} | D_{g,1} = 1, D_{g,2} = 1)$, if a group is more likely to be treated if her treatment effect is initially high.

definition. Estimators of $\underline{\sigma}_{fe}$ and $\underline{\sigma}_{fd}$ (or $\underline{\sigma}_{fe}$ and $\underline{\sigma}_{fd}$) can then be used to determine which of $\widehat{\beta}_{fe}$ or $\widehat{\beta}_{fd}$ is more robust to heterogeneous treatment effects.

Finally, and similarly to the result shown in Corollary 2 for β_{fe} , β_{fd} is equal to δ^{TR} under common trends and the following assumption:

Assumption 8 (*w_{fd} uncorrelated with $\widetilde{\Delta}$*) $E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{fd,g,t} - 1) (\Delta_{g,t} - \Delta^{TR}) \right] = 0$.

Note that under the common trends assumption, one can jointly test Assumption 8 and Assumption 7, the assumption that the weights attached to β_{fe} are uncorrelated with the $\Delta_{g,t}$ s: if $\widehat{\beta}_{fe}$ and $\widehat{\beta}_{fd}$ are significantly different, at least one of these two assumptions must fail. In the second application we revisit in Section 6, $\widehat{\beta}_{fe}$ and $\widehat{\beta}_{fd}$ are significantly different.

4 An alternative estimator

In this section, we show that it is possible to estimate a well-defined causal effect even if treatment effects are heterogeneous across groups or over time. Let

$$\delta^S = E \left[\frac{1}{N_S} \sum_{(i,g,t): t \geq 2, D_{g,t} \neq D_{g,t-1}} [Y_{i,g,t}(1) - Y_{i,g,t}(0)] \right],$$

with $N_S = \sum_{(g,t): t \geq 2, D_{g,t} \neq D_{g,t-1}} N_{g,t}$. δ^S is the ATE of all switching cells. In staggered adoption designs, δ^S is the average of the treatment effect at the time when a group starts receiving the treatment, across all groups that become treated at some point.

We now show that δ^S can be unbiasedly estimated by a weighted average of DID estimators. This result holds under the following supplementary assumptions.

Assumption 9 (*Strong exogeneity for $Y(1)$*) For all $(g, t) \in \{1, \dots, G\} \times \{2, \dots, T\}$, $E(Y_{g,t}(1) - Y_{g,t-1}(1) | D_{g,1}, \dots, D_{g,T}) = E(Y_{g,t}(1) - Y_{g,t-1}(1))$.

Assumption 9 is the equivalent of Assumption 4, for the potential outcome with treatment. It requires that the shocks affecting a group's $Y_{g,t}(1)$ be mean independent of that group's treatment sequence.

Assumption 10 (*Common trends for $Y(1)$*) For $t \geq 2$, $E(Y_{g,t}(1) - Y_{g,t-1}(1))$ does not vary across g .

Again, Assumption 10 is the equivalent of Assumption 5, for the potential outcome with treatment. It requires that between each pair of consecutive periods, the expectation of the outcome with treatment follow the same evolution over time in every group. Assumptions 9 and 10 ensure that one can reconstruct the potential outcome that groups leaving the treatment between

$t - 1$ and t would have experienced if they had remained treated. In staggered adoption designs, Assumption 9 and 10 are not necessary for identification, because no group leaves the treatment. Together, Assumptions 5 and 10 imply that the ATE follows the same evolution over time in every group: $E(\Delta_{g,t}) = \eta_t + \theta_g$.¹³ This still allows for heterogeneous treatment effects across groups and over time.¹⁴

Assumption 11 (*Existence of “stable” groups*) For all $t \in \{2, \dots, T\}$:

1. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 0$.
2. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 1, D_{g,t} = 0$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 1$.

The first point of the stable groups assumption requires that between each pair of consecutive time periods, if there is a “joiner” (i.e., a group switching from being untreated to treated), then there should be another group that is untreated at both dates. The second point requires that between each pair of consecutive time periods, if there is a “leaver” (i.e., a group switching from being treated to untreated), then there should be another group that is treated at both dates.

Notice that under Assumption 11, groups’ treatments are not independent, so Assumption 3 cannot hold. Accordingly, we replace Assumption 3 by Assumption 12 below. Assumption 12 requires that conditional on its own treatments, a group’s outcomes be mean independent of the other groups’ treatments. It is weaker than Assumption 3. Assumption 11 is necessary to show that our estimator is unbiased, but it is not necessary to show that it is consistent. Accordingly, in Section ?? of the Web Appendix, we show that our estimator is consistent under Assumption 3. For every $g \in \{1, \dots, G\}$, let $\mathbf{D}_g = (D_{1,g}, \dots, D_{T,g})$.

Assumption 12 (*Mean independence between a group’s outcome and other groups treatments*) For all g and t , $E(Y_{g,t}(0)|\mathbf{D}) = E(Y_{g,t}(0)|\mathbf{D}_g)$ and $E(Y_{g,t}(1)|\mathbf{D}) = E(Y_{g,t}(1)|\mathbf{D}_g)$.

We can now define our estimator. For all $t \in \{2, \dots, T\}$ and for all $(d, d') \in \{0, 1\}^2$, let

$$N_{d,d',t} = \sum_{g: D_{g,t}=d, D_{g,t-1}=d'} N_{g,t} \quad (3)$$

¹³ It should be possible to weaken Assumptions 9-10, in particular to account for dynamic effects where $\Delta_{g,t}$ may depend on $(D_{g,1}, \dots, D_{g,t-1})$. This would however introduce complications that are beyond the scope of this paper.

¹⁴ Imposing Assumptions 9 and 10 does not change the decompositions obtained in Theorems 1 and 2. $Y_{g,t}(1)$ is observed for all the treated (g, t) cells entering these decompositions, so those assumptions do not bring identifying information for those cells.

denote the number of observations with treatment d' at period $t - 1$ and d at period t . Let

$$\begin{aligned} \text{DID}_{+,t} &= \sum_{g:D_{g,t}=1,D_{g,t-1}=0} \frac{N_{g,t}}{N_{1,0,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=D_{g,t-1}=0} \frac{N_{g,t}}{N_{0,0,t}} (Y_{g,t} - Y_{g,t-1}), \\ \text{DID}_{-,t} &= \sum_{g:D_{g,t}=D_{g,t-1}=1} \frac{N_{g,t}}{N_{1,1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=0,D_{g,t-1}=1} \frac{N_{g,t}}{N_{0,1,t}} (Y_{g,t} - Y_{g,t-1}). \end{aligned}$$

Note that $\text{DID}_{+,t}$ is not defined when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 0$, or no group such that $D_{g,t} = 0, D_{g,t-1} = 0$. In such instances, we let $\text{DID}_{+,t} = 0$. Similarly, let $\text{DID}_{-,t} = 0$ when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 1$ or no group such that $D_{g,t} = 0, D_{g,t-1} = 1$. Finally, let

$$\text{DID}_M = \sum_{t=2}^T \left(\frac{N_{1,0,t}}{N_S} \text{DID}_{+,t} + \frac{N_{0,1,t}}{N_S} \text{DID}_{-,t} \right).$$

Theorem 3 *If Assumptions 1, 2, 4, 5, and 9-12 hold, $E[\text{DID}_M] = \delta^S$.*

In Section ?? of the Web Appendix, we also show that when G goes to infinity, DID_M is a consistent and asymptotically normal estimator of δ^S . The DID_M estimator is computed by the `fuzzydid` and `did_multipligt` Stata packages.

Here is the intuition underlying Theorem 3. $\text{DID}_{+,t}$ compares the evolution of the mean outcome between $t - 1$ and t in two sets of groups: the joiners, and those remaining untreated. Under Assumptions 4 and 5, $\text{DID}_{+,t}$ estimates the joiners' treatment effect. Similarly, $\text{DID}_{-,t}$ compares the evolution of the outcome between $t - 1$ and t in two sets of groups: those remaining treated, and the leavers. Under Assumptions 9 and 10, it estimates the leavers' treatment effect. Finally, DID_M is a weighted average of those DID. Note that in staggered designs, there are no groups whose treatment decreases over time, so DID_M is only a weighted average of the $\text{DID}_{+,t}$ estimators. Note also that one can separately estimate the joiners' and the leavers' treatment effect, by computing separately weighted averages of the $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ estimators. The former estimator only relies on Assumptions 4 and 5, while the latter only relies on Assumptions 9 and 10.

DID_M is related to two other estimators. First, it is related to the Wald-TC estimator in point 2 of Theorem S1 in the Web Appendix of de Chaisemartin and D'Haultfœuille (2018), but the weighting of $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ therein differs. As a result, DID_M estimates Δ^S under weaker assumptions. DID_M is also related to the multi-period DID estimator in Imai and Kim (2018). However, the multi-period DID estimator is a weighted average of the $\text{DID}_{+,t}$, so it does not estimate the leavers' treatment effect, and applies to a smaller population. Besides, Imai and Kim (2018) do not establish the properties of their estimator. Finally, they do not generalize it to non-binary treatments, something we do in Section ?? of the Web Appendix.